

Q1

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Consider the following system of linear equations.

$$4x + y + z = 6$$

$$x + 3y + z = 5$$

$$x + y + 2z = 4$$

Soln

① Augmented matrix, Echelon form, and Backward prog.

Step-1 form Aug. matrix.

$$[A|b] = \left[\begin{array}{ccc|c} 4 & 1 & 1 & 6 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & 2 & 4 \end{array} \right]$$

Step-2 Reduce to row echelon form.

Swapping R_1 and R_2 to get Leading 1.

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 5 \\ 4 & 1 & 1 & 6 \\ 1 & 1 & 2 & 4 \end{array} \right]$$

Now perform $\dots R_2 \leftarrow R_2 - 4R_1$ and $R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 5 \\ 0 & -11 & -3 & -14 \\ 0 & -2 & 1 & -1 \end{array} \right] \xrightarrow[\substack{\text{Normalize } R_2 \text{ to} \\ \text{get a Leading 1}}]{R_2 \leftarrow \pm \frac{1}{11} R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 5 \\ 0 & 1 & 3/11 & 14/11 \\ 0 & -2 & 1 & -1 \end{array} \right]$$

Now $R_3 \leftarrow R_3 + 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 5 \\ 0 & 1 & 3/11 & 14/11 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

Q1 (a) Continued
Step-3 . Backward Substitutions

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The matrix corresponds to system.

$$x + 3y + z = 5$$

$$y + \frac{3}{11}z = \frac{14}{11}$$

$$z = 1$$

Now substitute $z = 1$ into the second equation

$$y + \frac{3}{11}(1) = \frac{14}{11} \Rightarrow y = \frac{14}{11} - \frac{3}{11} = \frac{11}{11} = 1$$

Substitute $y = 1$ and $z = 1$ into the first equation

$$x + 3(1) + 1(1) = 5 \Rightarrow x + 4 = 5 \Rightarrow x = 1$$

Hence the solution to the system is

$$\boxed{x=1, y=1, z=1}$$

Q2

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Soln (b) Eigen decomposition.

The coefficient matrix A is.

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

To use eigen decomposition to solve $Ax = b$.

$\Lambda \rightarrow$ is diagonal matrix of eigenvalues. We need to find the eigenvalues & eigenvectors of A .

Such that $A = P \Lambda P^{-1}$, the solution is then $x = P \Lambda^{-1} P^{-1} \cdot b$.

Eigen values of A are $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 6$

Corresponding eigenvectors are

$$v_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

matrix P is formed by eigen vectors as columns.

form P and Λ . $P = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

Compute $P^{-1} = \frac{1}{\det(P)} \text{adj}(P)$, The determinant is 3

$$P^{-1} = \frac{1}{3} \begin{bmatrix} -1 & 1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

Compute $\Lambda^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/6 \end{bmatrix}$

Solving for $x = P \Lambda^{-1} P^{-1} b$.

$$x = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/6 \end{bmatrix} \frac{1}{3} \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix}$$

Q1 (b) Continued.

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$$x = \frac{1}{3} \begin{bmatrix} -1 & -1/2 & 1/6 \\ 0 & 1/2 & 1/6 \\ 1 & 0 & 1/6 \end{bmatrix} \begin{bmatrix} -1/6 - 1/5 + 2/4 \\ -1/6 + 2/5 - 1/4 \\ 1/6 + 1/5 + 1/4 \end{bmatrix} \quad (\text{MFML})$$

$$= \frac{1}{3} \begin{bmatrix} -1 & -1/2 & 1/6 \\ 0 & 1/2 & 1/6 \\ 1 & 0 & 1/6 \end{bmatrix} \begin{bmatrix} -3 \\ 0 \\ 15 \end{bmatrix}$$

$$x = \frac{1}{3} \begin{bmatrix} -1(-3) - 1/2(0) + 1/6(15) \\ 0(-3) + 1/2(0) + 1/6(15) \\ 1(-3) + 0(0) + 1/6(15) \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 + 2.5 \\ 2.5 \\ -3 + 2.5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 5.5 \\ 2.5 \\ -0.5 \end{bmatrix}$$

Q1 (c) Chokery decomposition:

The coefficient matrix, A is symmetric and positive definite, we can factorize A as:
 $A = LL^T$, where L is lower triangular matrix.

we solve $AX = b$ by solving two simpler systems:

$Ly = b$ for y (forward substitution)

then $L^T x = y$ for x (backward substitution)

plus $A = LL^T$.

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \begin{bmatrix} L_{11} & L_{12} & L_{13} \\ 0 & L_{22} & L_{23} \\ 0 & 0 & L_{33} \end{bmatrix}$$

$$\text{formula for } L_{ij} = \sqrt{A_{ij} - \sum_{k=1}^{j-1} L_{ik}^2}$$

$$L_{ij} = L_{ji} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk} \right) \text{ for } i > j$$

Q1 (c) continued

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for $j=1$

$$L_{11} = \sqrt{A_{11}} = \sqrt{4} = 2$$

$$L_{21} = A_{21}/L_{11} = 1/2$$

$$L_{31} = A_{31}/L_{11} = 1/2$$

for $j=2$

$$L_{22} = \sqrt{A_{22} - L_{21}^2} = \sqrt{3 - (1/2)^2} = \sqrt{3 - 1/4} = \sqrt{11}/2$$

$$L_{32} = \frac{1}{L_{22}} (A_{32} - L_{31}L_{21}) = \frac{2}{\sqrt{11}} (1 - (1/2)(1/2)) = \frac{2}{\sqrt{11}} (1 - 1/4) = \frac{2}{\sqrt{11}} (3/4) = \frac{3}{2\sqrt{11}}$$

for $j=3$

$$L_{33} = \sqrt{A_{33} - (L_{31}^2 + L_{32}^2)} = \sqrt{2 - ((1/2)^2 + (\frac{3}{2\sqrt{11}})^2)} =$$

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1/2 & \sqrt{11}/2 & 0 \\ 1/2 & 3/(2\sqrt{11}) & \sqrt{17}/11 \end{bmatrix} = \sqrt{2 - (1/4 + 9/44)} = \sqrt{17}/11$$

Now solve $Ly = b$

$$\begin{bmatrix} 2 & 0 & 0 \\ 1/2 & \sqrt{11}/2 & 0 \\ 1/2 & 3/(2\sqrt{11}) & \sqrt{17}/11 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix}$$

$$2y_1 = 6 \Rightarrow y_1 = 3$$

$$1/2 y_1 + \sqrt{11}/2 y_2 + \sqrt{17}/11 y_3 = 4$$

$$y_2 = 7/\sqrt{11}$$

$$1/2 y_1 + 3/(2\sqrt{11}) + \sqrt{17}/11 y_3$$

$$y_3 = \sqrt{17}/11$$

So finally

$$y = \begin{bmatrix} 3 \\ 7/\sqrt{11} \\ \sqrt{17}/11 \end{bmatrix}$$

Q1 (C) Continued

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$Lx = y$ for x (backward substitution).

$$\begin{bmatrix} 2 & 1/2 & 3/2 \\ 0 & \sqrt{11}/2 & 3/(2\sqrt{11}) \\ 0 & 0 & \sqrt{17/11} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 7/\sqrt{11} \\ \sqrt{17/11} \end{bmatrix}$$

$$\sqrt{17/11} \cdot x_3 = \sqrt{17/11} \Rightarrow x_3 = 1$$

$$\sqrt{11}/2 x_2 + 3/(2\sqrt{11}) x_3 = 7/\sqrt{11}$$

$$\Rightarrow \sqrt{11}/2 x_2 + 3/(2\sqrt{11}) (1) = 7/\sqrt{11}$$

$$\Rightarrow \sqrt{11}/2 x_2 = 7/\sqrt{11} - 3/(2\sqrt{11}) = (14-3)/(2\sqrt{11})$$

$$\Rightarrow x_2 = (14-3)/(2\sqrt{11}) = 11/(2\sqrt{11}) = \sqrt{11}/2$$

$$\# 2x_1 + 1/2 x_2 + 1/2 x_3 = 3$$

$$2x_1 + 1/2 x_1 + 1/2 x_1 = 3$$

$$x_1 = 1. \quad \text{Hence } x = 1, y = 1, z = 1$$

Q1 (a) Comparison of above method

Augmented

$O(n^3)$

1

2. most efficient

Eigen decomposition

① $O(n^3)$ ($O(n^4)$)
more complex

② Least efficient

Cholesky

① $O(n^3)$ but more or twice efficient than AM for symmetric +ve definite matrix

② Efficient, but slightly less than AM elimination for single solution

Q2

Define an inner product

 $\mathbb{R}^4$ by

$$\langle x, y \rangle = x^T M y,$$

where $M = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}, x, y \in \mathbb{R}^4$

Consider the vectors,

$$u = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, w = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

Soln.

Verify inner product properties

To verify that $\langle x, y \rangle = x^T M y$ defines an inner product on $\mathbb{R}^4$, we must check the following properties for all $u, v, w \in \mathbb{R}^4$ and $\alpha \in \mathbb{R}$.

$\Rightarrow$ Symmetry: $\langle v, w \rangle = \langle w, v \rangle$, matrix must be

Since given M is symmetric but need to prove inner product: linearity, symmetry & positive definiteness

1. Linearity in the first argument

$$\begin{aligned} \langle au + bv, w \rangle &= (au + bv)^T M w = (a u^T + b v^T) M w \\ &= a u^T M w + b v^T M w = a \langle u, w \rangle + b \langle v, w \rangle \end{aligned}$$

2. Symmetry

$\langle x, y \rangle = x^T M y$; the transpose of this scalar is

$$(x^T M y)^T = y^T M^T x$$

We need for symmetry $M = M^T$

$$M^T = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Since $M = M^T$, the property $\langle x, y \rangle = \langle y, x \rangle$

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Q2 (a) Continued

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③ positive definiteness

$$(x, x) = x^T M x \geq 0, \text{ and } (x, x) = 0.$$

$$\text{It and any } x \cdot x = 0.$$

This requires the
matrix M is to be
positive definite

$$D_1 = \det(1) = 2 > 0$$

$$D_2 = \det \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = 6 - 1 = 5 > 0$$

$$D_3 = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4 \det \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = 4 \times 5 = 20 > 0$$

$$D_4 = \det(M) = \det \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \det \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} = 2 \times 3 = 6 > 0$$

Since all leading principal minors are positive, M is
positive definite.

Hence $\langle \cdot, \cdot \rangle$ is a valid inner product on $\mathbb{R}^4$
because it satisfies the properties of
linearity, symmetry, & positive definiteness.

Q2 (b) Soln:-

Compute $\langle u, v \rangle$, $\langle u, w \rangle$ and $\langle v, w \rangle$

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$$\Rightarrow \langle u, v \rangle = u^T M v = [1 \ 2 \ 0 \ 1] \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad (\text{mfml})$$

$$\langle u, v \rangle = [1 \ 2 \ 0 \ 1] \begin{bmatrix} 2(0) - 1(1) + 0 + 0 \\ -1(0) + 3(1) + 0 + 0 \\ 0 + 0 + 4(1) + 1(1) \\ 0 + 0 + 1(1) + 2(1) \end{bmatrix} = [1 \ 2 \ 0 \ 1] \begin{bmatrix} -1 \\ 3 \\ 5 \\ 3 \end{bmatrix}$$

$$\langle u, v \rangle = 1(-1) + 2(3) + 0(5) + 1(3) = -1 + 6 + 0 + 3 = 8$$

$$\Rightarrow \langle u, w \rangle = u^T M w = [1 \ 2 \ 0 \ 1] \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

$$= [1 \ 2 \ 0 \ 1] \begin{bmatrix} 2(-1) - 1(0) + 0 + 0 \\ -1(-1) + 3(0) + 0 + 0 \\ 0 + 0 + 4(2) + 1(0) \\ 0 + 0 + 1(-1) + 2(0) \end{bmatrix} = [1 \ 2 \ 0 \ 1] \begin{bmatrix} -2 \\ 1 \\ 8 \\ -1 \end{bmatrix}$$

$$\langle u, w \rangle = 1(-2) + 2(1) + 0(8) + 1(-1) = -2 + 2 + 0 - 1 = -1$$

$$\Rightarrow \langle v, w \rangle = v^T M w = [0 \ 1 \ 1 \ 1] \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

$$= [0 \ 1 \ 1 \ 1] \begin{bmatrix} 2(1) - 1(0) + 0 + 0 \\ -1(1) + 3(0) + 0 + 0 \\ 0 + 0 + 4(2) + 1(0) \\ 0 + 0 + 1(2) + 2(0) \end{bmatrix} = [0 \ 1 \ 1 \ 1] \begin{bmatrix} 2 \\ -1 \\ 8 \\ 2 \end{bmatrix}$$

$$= 0(2) + 1(-1) + 1(8) + 1(2) = 0 - 1 + 8 + 2 = 9$$

Hence inner products are

$$\begin{cases} \langle u, v \rangle = 8 \\ \langle u, w \rangle = -1 \\ \langle v, w \rangle = 9 \end{cases}$$

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Q2 (c). Compute the norms.

$$\|u\|, \|v\|, \|w\|$$

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Soln

The norm is defined as

$$\|x\| = \sqrt{(x, x)}$$

$$\rightarrow \|u\| \rightarrow$$

$$(u, u) = u^T M_u = [1 \ 2 \ 0 \ 1] \begin{bmatrix} 2 \\ -1 \\ 8 \\ 2 \end{bmatrix}$$

from calculation in part (b)

$$M_u = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(1) - 1(2) + 0 + 0 \\ -1(1) + 3(2) + 0 + 0 \\ 0 + 0 + 4(0) + 1(1) \\ 0 + 0 + 1(0) + 2(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 1 \\ 2 \end{bmatrix}$$

$$(u, u) = u^T (M_u) = [1 \ 2 \ 0 \ 1] \begin{bmatrix} 0 \\ 5 \\ 1 \\ 2 \end{bmatrix} = 1 \times 0 + 2 \times 5 + 0 \times 1 + 1 \times 2 = 12$$

$$\|u\| = \sqrt{12} = 2\sqrt{3}$$

$$\rightarrow \|v\|$$

$$M_v = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 5 \\ 3 \end{bmatrix}$$

from (b)

$$(v, v) = v^T (M_v) = [0 \ 1 \ 1 \ 1] \begin{bmatrix} -1 \\ 3 \\ 5 \\ 3 \end{bmatrix} = 0(-1) + 1(3) + 1(5) + 1(3) = 11$$

$$\|v\| = \sqrt{11}$$

$$\rightarrow \|w\|$$

$$M_w = \begin{bmatrix} 2 \\ -1 \\ 8 \\ 2 \end{bmatrix}$$

from (b)

$$(w, w) = w^T M_w = [1 \ 0 \ 2 \ 0] \begin{bmatrix} 2 \\ -1 \\ 8 \\ 2 \end{bmatrix} = 1(2) + 0(-1) + 2(8) + 0(2) = 18$$

$$\|w\| = \sqrt{18} = 3\sqrt{2}$$

Hence Norms are

$$\|u\| = \sqrt{12}, \|v\| = \sqrt{11}, \text{ and } \|w\| = \sqrt{18}.$$

Q2 (2) Show vectors u, v & w
are linearly independent

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Soln A set of vectors is linearly indept.

When $c_1u + c_2v + c_3w = 0$, $c_1 = c_2 = c_3 = 0$.

We can form matrix U with these vectors as
columns and check its rank/det.

$$U = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix}$$

Since 4×3 matrix, we check rank. We can
use row reduction or check the det of any 3×3 submatrix.

Consider the top 3×3 submatrix U'

$$U' = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \Rightarrow \det(U) = 1(1 \times 2 - 0 \times 1) - 0(2 \times 2 - 0 \times 0) + 1(2 \times 1 - 1 \times 0) = 4$$

Since determinant is non-zero.

the columns of U' are linearly indept. Hence u, v & w also

Q2 (e) Gram-Schmidt process with (u, v, w) to obtain an orthonormal set.

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Solⁿ using inner product (u, v, w)
 $(x, y) = x^T \cdot y$ orthonormal set (q_1, q_2, q_3)

S-1 normalize 1st vector
 $\|u\| = 3\sqrt{2}$ $q_1 = \frac{u}{\|u\|} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

S-2 orthogonalize v wrt u to get e_2 , then normalize e_2

$$e_2 = v - \frac{(v, u)}{\|u\|^2} u \quad (\text{using } (v, u) = 8, \text{ and } \|u\|^2 = 18)$$

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{8}{18} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{4}{9} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} -4 \\ 9 \\ 9 \end{bmatrix} = \frac{\sqrt{47}}{3}$$

$$q_2 = \frac{e_2}{\|e_2\|} = \frac{3}{\sqrt{47}} \cdot \frac{1}{9} \begin{bmatrix} -4 \\ 9 \\ 9 \end{bmatrix} = \frac{1}{3\sqrt{47}} \begin{bmatrix} -4 \\ 9 \\ 9 \end{bmatrix}$$

S-3 orthogonalize w wrt to q_1 and q_2 to get e_3 .

$$e_3 = w - (w, q_1)q_1 - (w, q_2)q_2$$

$$\text{or } e_3 = w - \frac{(w, u)}{\|u\|^2} u - \frac{(w, e_2)}{\|e_2\|^2} e_2$$

using previously calc'd $(w, u) = 2, (w, e_2) = 73/9$

$$e_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{2}{18} u - \frac{73/9}{47/9} e_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{9} u - \frac{73}{47} e_2$$

$$e_3 = \frac{1}{423} \begin{bmatrix} 356 \\ -159 \\ 394 \\ -73 \end{bmatrix} \quad q_3 = \frac{e_3}{\|e_3\|} = \frac{\sqrt{423}}{\sqrt{442}} \begin{bmatrix} 1 \\ 1 \\ 423 \\ -73 \end{bmatrix}$$

Hence

$$q_1 = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, q_2 = \frac{1}{3\sqrt{47}} \begin{bmatrix} -4 \\ 1 \\ 9 \\ 5 \end{bmatrix}, q_3 = \frac{\sqrt{423}}{423\sqrt{42}} \begin{bmatrix} 356 \\ -159 \\ 394 \\ -73 \end{bmatrix}$$